

Plan update for Alex Gates — status, model duality, and path ahead (full)

Date: 2026-06-11

From: Charles Levine (draft for your edit)

One-page summary: [20260611_Brief_for_Alex_Gates_brief.md](#) — Charles converts to PDF when ready (`pdf_styles_one_page.css` via `convert_single_md_to_pdf.sh`).

Purpose: Keep you aligned on project status, the main modeling tension, and **what I’m going with** for a **Summer–Fall 2026** core manuscript draft/submission. This is not a decision memo — I’m informing you of my sequencing choices after a cross-project COMPASS review.

1. Where we stand

We have a **credible three-setting empirical story**: inverted-U in advancement vs **leave-one-out (LOO) peer-pool quality** in Army (anchor) and college basketball (replication), with academia (tenure) as a **preliminary third panel (stage 9 bins)**. The scientific bottleneck is no longer “find the curve” but **converge on claim language, minimal generative mechanism, and 2–3 testable predictions** for a Wang-style paper — without letting Army 525/UIC work, network extensions, or generative perfectionism block the draft.

2. What is solid

Area	Status
Empirical inverted-U (LOO axis)	Army · Basketball (<code>poolq_loo</code> / draft rate) · Tenure preliminary
Wang empirical ladder (basketball)	Bins → LPM → logit in <code>538</code> ; L^* reported
Minimal score equation	$S_i = A_i - \lambda \cdot L_{C,LOO}$ implemented in <code>538D</code> CELL 10
Ability-only generative null	Confirmed — congestion term needed for nonlinear readout
Generative POC (one axis)	Inverted-U vs pool mean (<code>team_mean</code> , includes self) under current top- K knobs
Tenure measurement	End-to-end stage 9 on existing data per your April direction; sample loss logged

3. The model duality (context)

Two related programs have been running in parallel. They are **not** the same estimand unless we nest them explicitly in prose.

A. Empirical stylized fact (cross-domain)

- **Axis: LOO pool quality** — mean teammate performance **excluding self** (`poolq_loo` in basketball; analogous objects in Army and tenure).
- **Finding:** Draft / promotion / tenure rates rise through mid-quality peer environments, then fall in the elite tier (**inverted-U**).
- **Role in paper:** Wang Rung 1 — the replicated phenomenon.

B. Generative minimal model (Alex score + assignment)

- **Score:** $S_i = A_i - \lambda \cdot L_{C,LOO}$ — own ability minus **LOO congestion** (viable peers excluding self).
- **Current readout:** Inverted-U appears when success is conditioned on **pool mean** (whole-roster average, **not** LOO).
- **On the empirical axis:** With the same knobs, generative readout vs **LOO pool quality** is **mostly decreasing**, not inverted-U.
- **Role in paper:** Wang Rung 2 — proof that finite distinction / congestion can bend advancement; **partial** match to Rung 1 unless axis language is careful.

C. Decomposition / predictions track

- **Structure:** Local environment as one object: $L_{net} = B(\cdot) - D(\cdot)$ — **benefit (development)** minus **constraint (congestion / distinction cost)**.
- **Empirical mechanism columns** (`congestion_quality` , LOO dispersion, minutes, etc.) intended as **diagnostics of B vs D**, not a second mechanism.
- **Tension:** Prior guidance sounded like “**Alex model for curves, multivariate empirical for predictions**” — workable only if nested in **one** ontology; otherwise it reads as **two models**. My plan below commits to **one story in writing**.

D. Terminology lock

Quantity	Name rule
Excluding self	Must say LOO (e.g. LOO pool quality, LOO congestion)
Whole roster including self	Pool mean / <code>team_mean</code> — not LOO

The program is **not** stuck on whether the empirical U exists. The work ahead is making the **generative object**, **empirical decomposition**, and **conditioning axis** read as **one scientific story** in the manuscript — with honest language about the LOO generative gap.

4. What I’m going with (near-term)

My sequencing choices

Area	My call
Timeline	Core manuscript draft/submission Summer–Fall 2026
Generative path	Manuscript-first: draft on empirical LOO U + honest generative POC (pool-mean readout) + explicit two-row axis table ; not blocking on generative LOO-pool-quality bin-for-bin match before first draft
Mechanism prose	Alex score = constraint-leg POC ; empirical decomposition for predictions — one $L = B - D$ story in §3–§4 (SCOUT to supply nesting chain for VECTOR)
Tenure (Setting 3)	Stage 9 + limitations → start Setting 3 prose now; one Cell 12 Cox in parallel before submission; Fine–Gray deferred
Deferred	Army 525/TB-stratify · network extensions · full HPC sweeps · LOO generative bin-for-bin match

Manuscript layers (what blocks draft vs not)

Layer	Manuscript role	Blocks first draft?
Empirical LOO inverted-U (3 settings)	Main empirical contribution	No
Alex score + assignment (pool-mean readout)	Minimal generative POC + axis table	No
B–D nesting prose + mechanism-column predictions	§3 mechanism + §4 predictions	Yes — needs SCOUT/VECTOR write-up
Generative match on LOO pool quality	Ideal upgrade	No

Scientific north star (longer run — parallel, not gating)

I still think the cleanest end state is a **single decomposable generative model** that keeps the Alex score as the constraint leg (D), adds an explicit benefit/exposure channel (B) where needed, and reproduces an inverted-U on **LOO pool quality** — the same axis as the empirical stylized fact. I will pursue that **in parallel if time allows** (e.g. a bounded SCOUT nesting sprint). If it lands before submission, we upgrade §3; if not, we ship with limitations.

5. Sequenced work (next ~6–10 weeks)

Week 1–2	Lock claim language · SCOUT exports (empirical Fig 2 + generative + axis table) · PEER Cell 12 Cox in parallel (not blocking first VECTOR draft) · CODA manuscript figures + estimand-checked captions
Week 2–4	VECTOR drafts Wang-structure manuscript (empirical triad → minimal model → predictions) · SCOUT supplies nesting chain (see §6)
Parallel	Optional SCOUT work toward LOO generative nesting (north star) · Tenure Cox table before submission · Defer: Army 525/TB-stratify, network extensions, full HPC sweeps, Fine-Gray

6. Internal next step (SCOUT — for your awareness)

Charles routes to SCOUT: [3-Master_Plan/20260611_0911_COMPASS_to_SCOUT_model_coherence_questions.md](#)

Under the manuscript-first plan, SCOUT will supply:

1. **Nesting chain** — where the Alex score lives in $L_{\text{net}} = B - D$; mechanism-column map to B vs D.
2. **Exact sentence** VECTOR may use in §3 so reviewers do not read “two mechanisms.”
3. **Deliverables** — axis table, frozen score equation, figure paths, one prediction from the same story.
4. **Single next coding task** on the manuscript path (not an open-ended rebuild first).

This follows your April–May themes (theory ≠ minimal model; quadratic as diagnostic; assortative pools + local comparison) as I understand them. If you see a conflict with that framing, I'd like to know.

7. Tenure (Setting 3) — how I’m applying your April direction

- **Stage 9 binned figure** + honest limitations: sufficient for VECTOR to **start** Setting 3 prose.
- **One basic Cell 12 Cox run** (mirror Army: z-scored LOO pool quality, quadratic, own-performance control) — **parallel before submission**, not a pre-draft blocker.
- **Fine-Gray**: deferred.
- Matches your “end-to-end on existing data, log sample loss, then move on” guidance as I’m interpreting it.

8. Where I'd value your perspective (optional)

Not asking you to choose a path — the table in §4 is my call. If anything below looks off, I'm happy to adjust:

- Whether **manuscript-first** with honest LOO generative limitations is acceptable for a core paper you'd stand behind, vs. needing LOO generative match pre-submission.
- How you'd **nest** "score for generative curve / decomposed empirical for predictions" in one Wang object — if that's still your operational framing.
- Whether a minimal B (development/exposure) term in sim is the right lever for the LOO axis, vs. primarily an estimand/conditioning issue.
- Whether **CELL 7 robustness**, **CELL 4D heterogeneity**, or **Army TB-stratify** should move up from the defer list.

9. Key repo pointers

Doc	Content
3- Master_Plan/PROJECT_STATUS_AND_NEAR_TERM_PLAN.md	Full near-term plan
3-Master_Plan/20260611_COMPASS_Initial_Review.md	Structured situational review
3-Master_Plan/SCOUT_report_to_COMPASS.md	Basketball / generative status
3-Master_Plan/PEER_report_to_COMPASS.md	Tenure status
3-Master_Plan/CODA_report_to_COMPASS.md	Army status
sports/documents/538_Cell10_Generative_Manual.md	CELL 10 implementation reference

Draft for Charles review and edit before sending to Alex.